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MILITARY SPECIFICATION

CORD ASSEMBLY, ELECTRICAL; CX-2151( )/U (RETRACTILE)

1. SCOPE

1.1 This specification covers Cord Assembly CX-2151( )/U. The electrical cord, which is used in the fabrication of the assembly, contains seven tinsel conductors insulated with a natural rubber compound, twisted together, covered with a styrene butadiene rubber (SBR) or a blend of SBR and natural rubber jacketing compound, and vulcanized into a helical form. The cord assembly consists of a length of retractile electrical cord having uncoiled portions of cord on each end partially stripped and terminated with solderless terminals. The electrical cord assembly is part of Handset H-60( )/U and is used with several different telephone sets, such as Telephone Set TA-312( )/PT.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-572	- Cords, Yarns, and Monofilaments; Organic Synthetic Fiber.
MIL-W-3795	- Wire, Electrical (Tinsel).

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MILITARY

MIL-I-3930	- Insulating and Jacketing Compounds, Electrical (For Cable, Cord and Wire).
*MIL-H-11996	- Handset H-60( )/PT, H-156( )/U, H-165( )/U.
*MIL-T-14358	- Telephone Set TA-312( )/PT.
MIL-C-45662	- Calibration System Requirements.
MIL-C-55442	- Cable Assemblies and Cord Assemblies, Packaging Of.

STANDARDS

FEDERAL

FED-STD-228	- Test Method Standard; Cable and Wire, Insulated; Methods of Testing.

MILITARY

MIL-STD-104	- Limits for Electrical Insulation Color.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

SIGNAL CORPS

SC-DL-19832	- Cord Assembly, Electrical CX-2151( )/U (-ft. -in.).

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*Supplied for information purposes only.

3. REQUIREMENTS

3.1 First article. The cords furnished under this specification shall be products which have been tested to and have passed the first article inspection requirements specified in 4.4. First article approval is valid only on the contract under which it is granted, unless extended by the government to another contract.

3.2 Construction. Cord Assembly, Electrical CX-2151( )/U shall be constructed in accordance with the drawings on Drawing and Data List SC-DL-19832 and shall meet the requirements of this specification. In case of a conflict between the specification and drawings, the specification shall govern.

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3.2.1 Construction details. The jacketed cord shall be helical in form with adjacent turns contiguous to form a helical cord, with a straight length of cord on each end tangent to the helical coil. The finished cord shall have a left-hand lay in its helical section. The helix shall have an overall diameter of not more than 1-3/6 inch. The helical portion of the cord shall be 9 inches ±3/4 inch long with the coils contiguous, unless otherwise specified. The finished cord shall have a smooth and uniform jacket.

3.3 Materials. The materials used in the construction of the cord covered by this specification shall be as specified herein. When a definite material is not specified, the best material commercially available for the purpose shall be used.

3.3.1 Conductors. Each conductor comprising the cord shall be tinsel wire conforming to MIL-W-3795, Type III.

3.3.2 Separator. Each conductor, prior to insulating, shall be closely covered with a separator yarn of polyester, Type PSTR, cellulose acetate, Type AR, or polyamide, Type P, per MIL-C-572. Vegetable and sulfonated oils shall not be applied as a lubricant to the textile yarns used in, or in contact with, the tinsel conductors.

3.3.2.1 Conductivity of separator yarns. The conductivity of the solution prepared from any part of the yarn removed from the conductor separator shall not exceed 50 micro-mhos per centimeter cube at 20°C, when tested as specified in 4.6.3.7.

3.3.3 Insulation. Each textile covered conductor shall be insulated with a natural rubber compound conforming to MIL-I-3930, Type IR, except that the insulation aging requirements shall conform to 3.3.3.1.

3.3.3.1 Insulation aging. The insulation over the conductors shall not exhibit any cracks when viewed with a magnifying glass of at least 3X magnification, after being subjected to the accelerated aging test specified in 4.6.3.8.

3.3.3.2 Dimensions of insulated conductors. The nominal diameter of each insulated conductor shall be 0.060 inches, with a maximum diameter of 0.070 inches. The insulation wall thickness shall be not less than 0.009 inches.

3.3.3.3 Insulation colors. The color of the insulation shall conform to MIL-STD-104, class 2. The colors of the individual insulated conductors in a core complement shall be in the following sequence:

1. White or natural
2. Black
3. Red
4. Green
5. Blue (light)
6. Yellow
7. Orange

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3.3.4 Cabling. Six of the insulated conductors shall be twisted with a right-hand lay around the remaining conductor, with a maximum length of lay of 1-5/8 inches.

3.3.5 Jacket. The cabled conductors shall be jacketed with a smooth, dense, uniform black elastomer free of surface irregularities, and conforming to 3.3.5.1. The jacket shall be free-stripping.

3.3.5.1 Jacket material properties. The jacket shall be a styrene butadiene rubber (SBR) compound or a blend of SBR and natural rubber, the blend not exceeding 25 percent natural rubber by weight. The vulcanized rubber jacket materials shall meet the requirements of MIL-I-3930, Type JS-L for SBR, and Type JR for a natural rubber, except as specified in Table I. The actual values shall be recorded and forwarded with first article sample test data. (See 3.1.) Unless otherwise indicated, specimens shall be taken from the straight section of the finished cord; for first article qualification, cords may be made up with extra-long straight sections.

Table I. Properties of jacket material.

Property	Value
Unaged	
Tensile strength, minimum psi	1,200
Ultimate elongation, minimum average percent	175
with no value less than, percent	150
Aged	
Oxygen bomb, 95 ±1 hour at 70°C	
Tensile strength, minimum psi	1,000
Elongation, average of measurements, minimum percent	100

3.3.5.2 Anti-deterioration agents. The rubber compound shall contain anti-oxidants, anti-ozonants, and sunproofing waxes of the types and amounts necessary to provide long term protection.

3.3.5.3 Dimensions of jacketed cords. After vulcanization into the construction specified in 3.2, the average diameter of the jacketed cord, measured on the straight length, shall be 0.265 ±0.010 inch. The average diameter shall be taken as the mean of the maximum and minimum diameter at any cross-section of the straight length of cord. The jacket thickness at any cross-section of the coiled cord shall be not less than 0.010 inch, and not less than 0.020 inch on any straight portion.

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3.4 Electrical requirements.

3.4.1 Dielectric withstanding voltage. The insulated conductors shall withstand, without breakdown, an applied voltage of 500 volts dc (350 volts ac, rms), when tested as specified in 4.6.2.1.

3.4.2 Insulation resistance. Immediately after the insulated conductors have been subjected to the dielectric withstanding voltage test, the insulation resistance shall be measured as specified in 4.6.2.2. The insulation resistance of each insulated conductor shall be not less than 100 megohms - 1,000 feet.

3.4.3 Continuity. Each conductor of the finished cord shall be continuous, when tested as specified in 4.6.2.3.

3.4.4 Contact resistance variation. The contact resistance variation between each solderless terminal and its conductor shall not exceed 0.06 ohm, when tested as specified in 4.6.2.4.

3.5 Physical properties.

3.5.1 Terminal pull-out. Each solderless terminal shall withstand a pull of 3 pounds, applied between the terminal and the tinsel conductor, without any pull-out of the terminal from the conductor, when tested as specified in 4.6.3.1.

3.5.2 Stay-hook. The stay-hook shall withstand a pull of 18 pounds, applied between the stay-hook and the jacketed cord, without any slippage of the stay-hook along the cord, when tested as specified in 4.6.3.2.

3.5.3 Grommet seal (to cord). The molding of the grommet to the cord shall present a seal that shall not allow more than one air bubble in 10 seconds to escape from the finished cord, when tested as specified in 4.6.3.3.

3.5.4 Grommet seal (to panel). The grommet shall seal a 0.375 ±0.001 inch diameter hole in a 0.340 ±0.001 inch thick metal panel without any air leakage between the grommet and the panel, when tested as specified in 4.6.3.4. Also, the grommet shall pass readily, without undue force, through a 0.400 ±0.001 inch diameter hole in a 0.340 ±0.001 inch thick metal panel.

3.5.5 Cord extensile and retractile characteristics. Prior to meeting any extensile and retractile requirements, the finished cord shall be conditioned as specified in 4.6.3.5. After conditioning, the cord shall be placed on a horizontal surface, and the coils shall be contiguous.

3.5.5.1 Cord extension. The cord shall extend to at least 5 times its retracted length when subjected to the test specified in 4.6.3.5.1. The contiguous coils of the cord shall separate freely and uniformly during extension and retraction.

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3.5.5.2 Retraction (after static vertical loading). After being tested in accordance with 4.6.3.5.2, the percentage of retraction of the helical portion of the cord shall be not more than 110 percent of its original length.

3.5.5.3 Retraction at -55°C. The retracted length of the helical portion of the cord shall be not more than 350 percent of its original retracted length, when tested as specified in 4.6.3.5.3. The jacket, insulation, and conductors shall be free from cracks or other defects when viewed at room temperature with magnifying glass of at least 3X magnification.

3.5.5.4 Retraction at 85°C. The retracted length of the helical portion of the cord shall be not more than 275 percent of its original retracted length, when tested as specified in 4.6.3.5.4. The jacket, insulation, and conductors shall be free from cracks or other defects when viewed at room temperature with a magnifying glass of at least 3X magnification.

3.5.6 Cold bend. After being subjected to the test specified in 4.6.3.6, the jacket and insulation from the straight portion of the cord shall be free from cracks, breaks, or other faults, when viewed at room temperature with a magnifying glass of at least 3X magnification.

3.6 Workmanship. Cords shall be constructed and finished in a thoroughly workmanlike manner in accordance with accepted high grade production techniques. The cords shall be a uniform and consistent product and shall be free from any defects which will adversely affect the serviceability of the product, such as lumps, kinks, splits, abrasions, scrapes, corroded surfaces, skin impurities and faulty extruded surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment and facilities. Inspection equipment and facilities shall be established and maintained in accordance with MIL-C-45662.

4.2 Classification of inspections. The examinations and tests of cords are classified as follows:

(a) Materials inspection. (See 4.3.)

(b) First article inspection. (See 4.4.)

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(c) Quality conformance inspection. (See 4.5.)

1. Inspection of product for delivery. (See 4.5.1.)
2. Inspection of preparation for delivery. (See 4.7.)

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in Table II, used in fabricating coiled cords, are in accordance with the applicable referenced specification or requirements prior to such fabrication.

Table II. Materials inspection.

Material	Requirement paragraph	Applicable specification
Conductor	3.3.1	MIL-W-3795, Type III
Separator	3.3.2	MIL-C-572,
Polyester		Type PSTR
Cellulose acetate		Type AR
Polyamide		Type P
Insulation	3.3.3	MIL-I-3930, Type IR
Jacket	3.3.5	MIL-I-3930, Type JS-L or JR

4.4 First article inspection. First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the government. (See 6.3.) First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. (See 3.1.)

4.4.1 Sample size and test routine. Twenty-five sample units shall be subjected to the tests specified in Tables III, IV and V. These sample units shall be of sufficient length to perform the tests specified.

4.4.2 Submission of data for approval. Test results (and samples, when specified) shall be forwarded to the contracting officer for approval.

4.4.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A, B, and C inspection. Delivery of products which have passed the groups A and B inspection shall not be delayed pending the results of the group C inspection.

4.5.1.1 Inspection lot. An inspection lot shall consist of all cords of the same type (see 3.2) produced under essentially the same conditions and submitted for inspection at the one time.

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4.5.1.2 Unit of product. The unit of product, for purposes of sampling, shall be each continuous length of cord as specified in the contract or order.

4.5.1.3 Sample. The sample shall consist of that number of randomly selected units of product required by the applicable sampling plan for the presented lot.

4.5.1.4 Sample units. A sample unit is a unit of product which forms part of the sample from which specimens are taken for inspection.

4.5.1.5 Specimen. A specimen may be the entire sample unit or any portion of the sample unit which is to be subjected to inspection.

4.5.2 Group A inspection. Group A inspection shall include the examinations and tests of Table III, subgroups I and II. Major and minor defects shall be as defined in MIL-STD-105.

4.5.2.1 Subgroup I. For subgroup I, the acceptable quality level (AQL) shall be as specified in Table III and the inspection level shall be level II in accordance with MIL-STD-105. Subgroup I tests may be performed in any order.

4.5.2.2 Subgroup II. For subgroup II, the AQL shall be as specified in Table III and the inspection level shall be level II in accordance with MIL-STD-105. Subgroup II tests shall be performed in the order shown in Table III.

Table III. Group A inspection.

Examination or test	Requirement paragraph	Test method paragraph	AQL (Percent defective) Major	AQL (Percent defective) Minor
Subgroup I				
Visual and dimensional				
Construction	3.2, 3.2.1	4.6.1	1.0% for the group combined	4.0% for the group combined
Materials	3.3			
Dimensions of insulated conductors	3.3.3.2			
Insulation colors	3.3.3.3			
Cabling	3.3.4			
Dimensions of jacketed cords	3.3.5.3			
Workmanship	3.6			
Subgroup II				
Electrical				
Dielectric withstanding voltage	3.4.1	4.6.2.1	1.0% for the group combined	*
Insulation resistance	3.4.2	4.6.2.2		
Continuity	3.4.3	4.6.2.3		
Contact resistance variation	3.4.4	4.6.2.4		

*All electrical defects are considered major.

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4.5.2.3 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.5.3 Group B inspection. This inspection, including sampling, shall conform to Table IV and to the procedures for small-sample inspection of MIL-STD-105, using special inspection levels. Group B inspection shall be performed on inspection lots that have passed group A inspection and on specimens selected from units of product that have been subjected to and met the group A inspection. The AQL shall be 6.5 (percent defective), and the special inspection level shall be S-2.

4.5.3.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the government.

4.5.3.2 Disposition of samples. Samples subjected to group B tests shall not be delivered on contract or order.

Table IV. Group B inspection.

Examination or test	Requirement paragraph	Test method paragraph
Terminal pull-out	3.5.1	4.6.3.1
Stay-hook	3.5.2	4.6.3.2
Grommet seal (to cord)	3.5.3	4.6.3.3
Grommet seal (to panel)	3.5.4	4.6.3.4
Cord extensile and retractile characteristics	3.5.5	4.6.3.5
Cord extension	3.5.5.1	4.6.3.5.1
Retraction (after static vertical loading)	3.5.5.2	4.6.3.5.2

4.5.4 Group C inspection. This inspection shall consist of the tests specified in Table V and shall be performed periodically as indicated in 4.5.4.1. Samples shall be selected from lots that have passed group A and B inspection.

4.5.4.1 Sampling for inspection. Two sample units for each group C inspection shall be selected each month through the life of the contract to assure representative results.

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Table V. Group C inspection.

Examination or test	Requirement paragraph	Test method paragraph
Conductivity of separator yarns	3.3.2.1	4.6.3.7
Insulation aging	3.3.3.1	4.6.3.8
Retraction at -55°C	3.5.5.3	4.6.3.5.3
Retraction at 85°C	3.5.5.4	4.6.3.5.4
Cold bend	3.5.6	4.6.3.6

4.5.4.2 Noncompliance. No failures shall be allowed in group C inspection. If a sample unit fails to pass group C inspection, the supplier shall take corrective action on the material or process or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the government, has been taken. After the corrective action has been taken, Group C inspection shall be repeated on additional sample units (all inspections or the inspection that the original sample failed, at the option of the government). Groups A and B inspection may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the cognizant inspection activity.

4.6 Test methods.

4.6.1 Visual and dimensional inspection. The finished cords shall be given a visual and dimensional inspection for conformance with the applicable requirements of Table III, subgroup I. Visual and dimensional defects shall be classified as major or minor in accordance with the definitions of MIL-STD-105.

4.6.2 Electrical tests.

4.6.2.1 Dielectric withstanding voltage (see 3.4.1). The finished cords shall be tested as specified in FED-STD-228, method 6111, except for the following:

(a) The test shall be performed on finished cords only.
(b) The immersion period shall be not less than 9 hours.
(c) One terminal shall be each conductor in turn, and the other terminal shall be all the remaining conductors tied together in electrical contact with the water.

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4.6.2.2 Insulation resistance (see 3.4.2). The insulation resistance of the finished cords shall be determined as specified in FED-STD-228, method 6031, except for the following:

(a) The test shall be performed on finished cords only.
(b) The immersion period shall be not less than 9 hours.
(c) The test voltage shall be not less than 100 volts nor more than 500 volts dc.
(d) The polarity of the conductor shall be maintained negative with respect to the water. One terminal shall be each conductor in turn, and the other terminal shall be all the remaining conductors tied together in electrical contact with one another and the water.
(e) If the measurement is made at a temperature lower than 15.6°C, the supplier shall correct the measured value of insulation resistance to the resistance at 15.6°C.

4.6.2.3 Continuity. Each conductor of the finished cord shall be tested for continuity using a maximum test voltage of 10 volts, and shall meet the requirements of 3.4.3.

4.6.2.4 Contact resistance variation. The conductor shall be stressed (wiggled) at the point of attachment of the conductor to its terminal. Each terminal shall be clamped so that it cannot move and the conductor shall be bent at the point of attachment to the terminal so that it is moved 90 degrees to each side of its normal position. The stressing (wiggling) shall be done in a vertical and a horizontal plane. The resistance variation may be measured with a bridge or by the use of an electronic voltmeter to measure the voltage variation, with a current of 0.30 to 0.50 ampere flowing through the conductor. The contact resistance variation shall meet the requirements of 3.4.4.

4.6.3 Physical tests.

4.6.3.1 Terminal pull-out. Each insulated conductor shall be gripped at a point at least one foot from the termination under test in such a manner that there will be no damage to the cord, nor slippage in the grip. The weight for the termination under test shall be applied in a manner to avoid a sudden jerk on the termination. After the force has been applied, the requirements of 3.5.1 shall be met.

4.6.3.2 Stay-hook. The stay-hook shall grip the cord in such a manner that there will be no damage to the cord jacket or insulation. The weight or calibrated pull for the stay-hook under test shall be applied gradually to avoid a sudden jerk on the stay-hook. After the force has been applied, the requirements of 3.5.2 shall be met.

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4.6.3.3 Grommet seal (to cord). The cord assembly shall be immersed in not more than 6 inches of water. While immersed, an air pressure of 5 psi shall be applied to one end of the cord. The leakage of air bubbles at the other end of the cord, if present, shall not exceed the requirements of 3.5.3. Leakage from the conductor shall be disregarded.

4.6.3.4 Grommet seal (to panel). One side of the panel containing a grommet (with cord attached) shall be immersed in water. An air pressure of 5 psi shall be applied to the exposed panel at the point of grommet insertion, to determine seal adequacy. The requirements of 3.5.4 shall be met.

4.6.3.5 Cord extensile and retractile characteristics. The cord shall be conditioned by extending the helical portion of the cord to 5 times its retracted length and allowing it to retract freely for a total of 6 cycles. After this conditioning, the cord shall meet the requirements of 3.5.5.

4.6.3.5.1 Cord extension. At room temperature, the cord, or helical portion thereof, shall be placed in a horizontal position. One end shall be fastened, with the helical portion contiguous. The length of the helical portion shall be measured, and a tensile force of 2 pounds applied to the free end of the cord to cause the cord to be extended. The ratio of extension shall meet the requirements of 3.5.5.1.

4.6.3.5.2 Retraction (after static vertical loading). At room temperature, the finished cord shall be secured at one end and hung vertically with a weight attached to the lower end, sufficient to extend the helical portion of cord to approximately 250 percent of its original horizontal retracted length. After 48 hours, the weight shall be removed and the cord placed on a horizontal surface. This surface shall be tapped to reduce frictional restraint. Thirty minutes after being placed on the horizontal surface, the helical portion of the cord shall be measured to determine percent retraction of original length. Percent retraction shall be calculated as follows:

$$\text{Percent retraction} = \frac{\text{Retracted length after extension}}{\text{Original retracted length}} \times 100$$

The measured value shall meet the requirements of 3.5.5.2.

4.6.3.5.3 Retraction at -55°C. A 6 inch helical portion of a finished cord with all coils contiguous shall be placed in a horizontal position within an environmental test chamber maintained at -55°C ±1°C, fastened at one end, and conditioned for not less than 20 hours prior to testing. If the helical portion of the finished cord is less than 6 inches long, the complete cord shall be used and measurements made of the helical portion. At the conditioning temperature, A force, not to exceed 20 pounds, shall be applied to the free end of the cord sufficient to extend the helical portion 5 times its retracted length within 5 seconds. While still at -55°C, the force shall be released and within 30 seconds, the helical portion of the specimen shall retract to not more than the value specified in 3.5.5.3, and shall meet the requirements specified therein. Frictional restraint shall be minimized by tapping the surface on which the cord rests.

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4.6.3.5.4 Retraction at 85°C. The test specimen shall be a finished cord. With the helical section extended to 3 times its retracted length, the cord shall be conditioned in a circulating air oven maintained at 85° +1°C for 20 hours. At the end of this period and while in the oven at 85°C, the helical section shall be extended further, to 5 times its retracted length, and returned to its original retracted length. The extension and retraction shall be repeated for a total of 3 cycles. The cord shall then be suspended vertically and measured for retraction at the end of 15 seconds, the percent retraction (calculated on the basis of original horizontal retracted length) shall meet the requirements specified in 3.5.5.4.

4.6.3.6 Cold bend. Specimens shall be taken from the straight portions of the cords. The specimens from any one cord shall be divided, one part for checking the finished cord as a whole (jacketed cord) and the other for checking insulation, for compliance with 3.5.6. The part selected for checking insulation shall have the jacket removed and each insulated conductor of the cord tested. The test specimens shall be attached to mandrels of diameters equal to the diameters of the specimens, and shall be suspended vertically with lower ends weighted sufficiently to keep the specimens taut and to permit bending them without handling. The mandrels and specimens shall be placed for at least 24 hours in a cold chamber maintained at -55° ±1°C. While at this temperature, the specimens shall be bent for 5 close turns around the mandrels at the rate of approximately 1 turn per second. After removal from the cold chamber, the specimens shall be examined for compliance with 3.5.6. Jacketed specimens shall also be stripped and the insulation of the conductors examined for compliance with the above requirement.

4.6.3.7 Conductivity of separator yarns. The separator yarns shall be subjected to the following test: One-half gram of material removed from the finished (but uninsulated) wire shall be boiled for 10 minutes in 60 to 70 cubic centimeters of distilled water which has a conductivity of not more than 5 micro-mhos per centimeter cube at 20°C. Enough distilled water shall then be added to make a volume of 100 cubic centimeters. The conductivity of this solution shall be measured with alternating current, and shall be determined as the average of two specimens removed from each sample unit. If the average thus determined exceeds the requirement specified in 3.3.2.1, the sample unit shall be considered defective. A conductivity cell may be used for this test.

4.6.3.8 Insulation aging. The individual insulated conductors of the finished cord shall be wound around a 1/8 inch mandrel at room temperature after being subjected to the accelerated aging test specified in FED-STD-228, Method 4011. The insulated conductors shall meet the requirements of 3.3.3.1.

4.7 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking shall be in accordance with MIL-C-55442. (See 6.1(d).)

6. NOTES

6.1 Intended use. Cord Assembly CX-2151( )/U is part of Handset H-60( )/U and is intended for use with various telephone sets (e.g. TA-312( )/PT).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required. (See 3.2.)
(c) Length and quantity of finished cords to be delivered.
(d) Level of packaging and level of packing delivered. (See 5.1.)

6.2.1 Indirect shipments. The preservation, packaging, packing and marking specified in Section 5 apply only to direct purchases by or direct shipments to the government, and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.2.2 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: CX-2151W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.3 First article inspection. Information pertaining to first article inspection of cords covered by this specification should be obtained from the procuring activity for the specific contracts involved. (See 3.1.)

6.3.1 First article inspection on repeat orders. First article inspection will not be required from a supplier who has manufactured the same type cord (same type of wire, number of conductors, number of shielded legs, etc.) on a previous order or contract within 2 years of the current order or contract, and which passed first article inspection at that time. This shall also apply when the only difference in the cord construction is in length.

Custodian:
Army - EL

Preparing Activity:
Army - EL

Project Nr. 5995-A087

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FO... CUT ALONG THIS LINE.)

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